

Physics-Informed Temporal Generative Adversarial Networks: A Potent Tool for Lithium Battery Data Generation

1st Chenyu Lin[#]

HDU-ITMO Joint Institute
Hangzhou Dianzi University
Hangzhou, China
173929702@qq.com

2nd Qiqi Chen[#]

School of Computer Science
Hangzhou Dianzi University
Hangzhou, China
22330109@hdu.edu.cn

3rd Guanyuan Pan

HDU-ITMO Joint Institute
Hangzhou Dianzi University
Hangzhou, China
panguanyuan@hdu.edu.cn

4th Zhuoxi Li

Zhuoyue Honors College
Hangzhou Dianzi University
Hangzhou, China
23270616@hdu.edu.cn

5th Zhongjie He^{*}

School of Artificial Intelligence
Hangzhou Dianzi University
Hangzhou, China
he_zj@hdu.edu.cn

6th Guoguo Ye

School of Computer Science
Hangzhou Dianzi University
Hangzhou, China
22061105@hdu.edu.cn

Abstract—Accurately predicting the aging process of lithium batteries is essential for battery health management and predictive maintenance. However, the scarcity of real aging data limits the reliability of existing models. To address this issue, this paper proposes a Physics-Informed Temporal Generative Adversarial Network (PITGAN)-based framework for generating multi-dimensional time-series battery data to support health assessment. The framework integrates temperature-aware feature embedding to dynamically incorporate thermal environment characteristics and utilizes a hybrid GRU-convolutional architecture to capture long-range temporal dependencies and local electrochemical features. Additionally, an improved GAN method is proposed, combining the conditional control capability of CGAN with the smooth optimization characteristics of WGAN-GP to enhance training stability and improve the authenticity of the generated data. The feature extraction module preserves the physical constraints of voltage-capacity curves, while the generator-discriminator structure enables joint modeling of multi-dimensional data distributions. The analysis of experimental results shows that the generated data closely resembles actual data in terms of distribution, with a voltage-capacity curve error of less than 1.8%. Furthermore, state-of-charge and state-of-health estimation models trained on the generated data achieve a mean absolute error below 3.2%.

Index Terms—Lithium Battery, Generative Adversarial Network, State of Charge, State of Health

I. INTRODUCTION

Driven by global energy transition, the rapid growth of the electric vehicle industry is fueled by lithium batteries, which offer high energy density and long cycle life, making them superior to traditional battery technologies [1].

Deep learning methods have been widely applied to estimate the State of Health (SOH) of batteries. Sedlařík et al. [2] developed a data-driven model to estimate the SOH of lithium batteries. Chae et al. [3] addressed environmental noise using a combination of denoising convolutional neural

networks through residual learning techniques. However, limited datasets can lead to model overfitting and data imbalance, which in turn affects the accuracy and robustness of battery fault detection [4], [5]. Consequently, augmenting datasets through data generation techniques and improving model performance has become a key focus of current research.

Currently, many researchers are dedicated to exploring battery data generation techniques to address the issue of data scarcity. Hu et al. [6] proposed a time-series enhancement model (TS-DCGAN), aiming to solve the data insufficiency problem by generating high-quality battery data. Iranipour et al. [7] employed data generation techniques using CGAN, successfully alleviating data scarcity in perovskite solar cells. Jin et al. [8] applied CGAN to generate synthetic data.

Physics-informed neural networks (PINN) and temporal modeling networks (GRU) generate data that closely resemble real-world values. However, each approach has inherent limitations when used independently. To overcome these challenges, this study proposes a physics-informed temporal generative adversarial network. The contributions of this research are summarized as follows:

- A hybrid PINN-GRU architecture that generates synthetic battery data.
- The incorporation of asymmetric gradient loss into the loss function to enhance training stability.
- A multi-scale convolutional feature extraction module that combines local transient signals and global aging trends at different scales using adaptive pooling.

The rest of the paper is organized as follows: [Section II](#) presents the method, [Section III](#) discusses the experiment, and [Section IV](#) provides the conclusion.

[#] These authors contributed to the work equally and should be regarded as co-first authors.

^{*} Corresponding author.

II. METHOD

A data generation network based on PITGAN (as shown in Fig. 1) is proposed for monitoring and prediction of SOH of lithium batteries, following these steps.

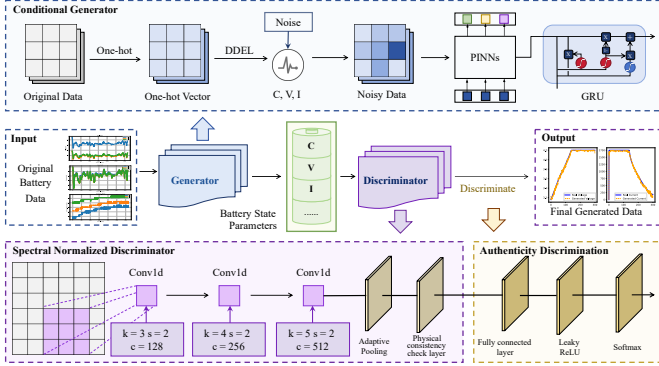


Fig. 1. Flowchart of the proposed PITGAN.

Step 1: Data Encoding and Conditional Generation

The original data is first encoded into a one-hot vector. The vector is then fed into a conditional generator, marking the starting point of the data processing pipeline.

Step 2: Noise Injection and DDEL Processing

Within the conditional generator, the one-hot encoded data is combined with injected noise. The noise is processed through a Deep Dynamic Evolutionary Learning (DDEL) module to generate noisy data, which serves as the input for the subsequent steps.

Step 3: Data Generation through the Generator

The noisy data is passed to the generator, where it is utilized to produce battery data, including time-series sequences of voltage (V), current (I), and capacity (C). This step is crucial for synthesizing data that mimics real-world conditions.

Step 4: Data Processing through GRU-Convolutional Architecture with Physical Consistency Check

In this step, the generated data is processed by a GRU to capture temporal dependencies and then passed through a GRU-Convolutional architecture. The architecture uses spectral normalization and convolutional layers with varying kernel sizes (k) and channel numbers (c) to extract relevant features. This is followed by an adaptive pooling layer and a physical consistency check layer that ensures the generated data adheres to known physical laws. The discriminator then analyzes the data, validating its authenticity against the expected behavior, providing the final generated data.

Step 5: Authenticity Discrimination

The discriminator produces a final output, which is first passed through a fully connected layer and then sequentially processed by a Leaky ReLU activation layer and a Softmax layer. These components classify the authenticity of the data, ensuring that the generated data adheres to the required physical consistency and authenticity.

Step 6: Authenticity Verification and Output

Finally, the generated data are compared to the real data to assess their authenticity. This step provides a verification

process, confirming that the generated data aligns with the expected physical behavior and satisfies authenticity criteria.

A. Conditional Generative Adversarial Network

Step 1: Conditional Generator

First, the input data is preprocessed, including One-Hot Encoding and data augmentation, to increase the diversity of the data. Then, the Deep Data Augmentation Learning (DDEL) method is applied, where the input data X_{battery} is transformed into noise-augmented data through a nonlinear mapping function:

$$z_{\text{noisy}} = f(X_{\text{battery}}, \epsilon) \quad (1)$$

where ϵ represents random noise, ensuring the diversity of the generated data. Next, the generator G_{φ} uses a multi-layer LSTM network to synthesize battery state sequences from the latent space:

$$\hat{X} = G_{\varphi}(z_{\text{noisy}}) \quad (2)$$

where $\hat{X} = \{\hat{I}_t, \hat{V}_t, \hat{C}_t\}$ is the synthesized battery state data. Skip connections are introduced into the LSTM to alleviate the vanishing gradient problem, allowing long-term dependency features to be preserved.

Step 2: Spectral-Normalized Discriminator

The task of the discriminator is to distinguish between real and generated data. A 1D convolutional architecture is adopted, combined with Spectral Normalization (SN) to stabilize adversarial training, constraining the Lipschitz constant of the convolutional layers to improve training stability. The network structure is as follows:

$$D_{\varphi}(x) = \text{Conv1D}_k(\text{LeakyReLU}(\text{Conv1D}_k(x))) \quad (3)$$

where k is the size of the convolution kernel, and SN constrains the maximum singular value of the convolution weight matrix W :

$$W_{\text{SN}} = \frac{W}{\sigma_{\max}(W)} \quad (4)$$

Step 3: Authenticity Discrimination Module

This module is used to map the output of the discriminator to a probability score, thus measuring the authenticity of the data. The feature extraction part of the discriminator first extracts features H_z through convolutional layers and then flattens them into a 1D vector h_{flat} . Next, a fully connected network processes the features, first performing a linear transformation and then applying another fully connected layer.

Finally, the Softmax function is used to calculate the authenticity probability distribution P . During training, a weighted cross-entropy loss function is used to address the class imbalance problem:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N w_i [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (5)$$

where w_i is the dynamically adjusted class weight, $y_i \in \{0, 1\}$ is the true label, and $\hat{y}_i$ is the discriminator's output. To further enhance the authenticity of the generated data, an adversarial loss feedback mechanism is used, allowing the generator to synthesize sequences closer to the real data:

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] - \mathbb{E}_{x \sim p_{\text{gen}}} [\log(1 - D(G(z)))]. \quad (6)$$

Through this approach, the CGAN generator can continuously optimize, making the synthetic data distribution closer to that of real data, thus improving the authenticity and diversity of the battery state data.

B. Physics-Informed Sequence Generation Network

Step 1: Hybrid PINN-Transformer Architecture

A Physics-Informed Neural Network (PINN) is integrated with a Transformer encoder to enforce electrochemical consistency. The Transformer utilizes multi-head self-attention to capture long-term dependencies:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) \cdot V \quad (7)$$

where Q, K, V are query, key, and value matrices derived from input embeddings.

Step 2: Physics-Constrained Loss Function

The generated sequences $\hat{X}$ are regularized by the governing equations of lithium batteries. For the Ohm-polarization model:

$$L_{\text{phy}} = \sum_{i=1}^T \left\| \hat{V}_t - [I_t R_{\text{int}} + U_{\text{ocv}}(\text{SOC}_t)] \right\|_2^2 + \lambda \left\| \frac{\partial \hat{Q}_t}{\partial t} - I_t \right\|_2^2 \quad (8)$$

where R_{int} is internal resistance, U_{ocv} is the open-circuit voltage function, and λ balances charge conservation constraints.

C. Combination of CGAN and PINNs

By combining the generative ability of CGAN with the physical constraints of PINN, it is possible to ensure data diversity while ensuring that the generated data aligns with the physical behavior of the battery, thereby enhancing the authenticity and reliability of the generated data. First, the battery state data (such as current, voltage, temperature, etc.) is preprocessed and transformed into latent space data, while incorporating random noise ϵ to generate the battery state sequence. The generator's objective is to learn the mapping from input data to latent space, generating the battery state sequence $\hat{X}$.

To ensure the physical consistency of the generated data, the output of the generator is not only based on traditional adversarial training but also incorporates PINN. PINN introduces the physical constraints of the battery, particularly the charge and discharge processes and the relationship between current and voltage, ensuring that the generated data aligns with the physical behavior of the battery. The generated data is regularized through physical constraint equations, with the Ohm-polarization model:

$$L_{\text{phy}} = \sum_{t=1}^T \left| \hat{V}_t - [I_t R_{\text{int}} + U_{\text{oc}}(\text{SOC}_t)] \right|^2 + \lambda \left| \frac{\partial \hat{Q}_t}{\partial t} - I_t \right|^2 \quad (9)$$

where $\hat{V}_t$ is the generated battery voltage, $\hat{I}_t$ is the battery current, R_{int} is the internal resistance of the battery, $U_{\text{oc}}(\text{SOC}_t)$ is the open-circuit voltage, and λ is the physical constraint coefficient, ensuring that the generated data complies with the electrochemical model of the battery.

III. EXPERIMENT RESULTS

A. Experiment Dataset

In this paper, we utilize three different battery datasets. Fig. 2 displays the raw data of three datasets.

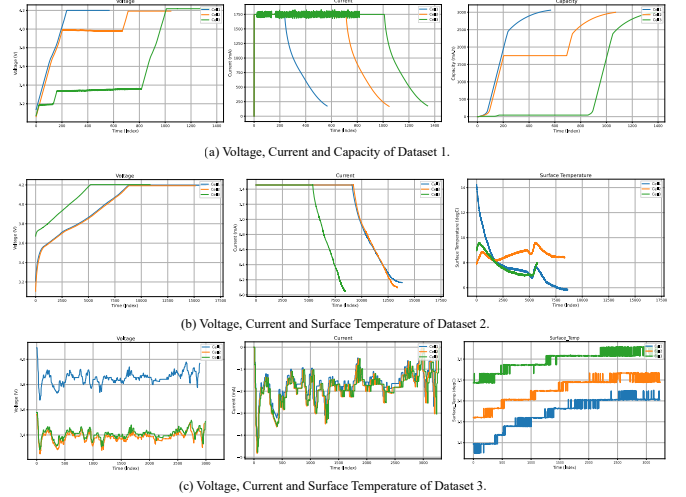


Fig. 2. The raw data of three datasets.

1. Dataset 1

These datasets contain cycling data for commercial 18650-type batteries, and we use the part from NCM batteries [9]. The C-rate was determined using the nominal capacity of the NCM cells, where 1C corresponds to 3.5 A. Visualization of voltage, current and capacity of this dataset is shown in Fig. 2 (a).

2. Dataset 2

For this dataset, we use the part generated from LG Chem INR21700-M50 (4850 mAh) [10], which has cathode chemistry of lithium-nickel-manganese-cobalt-oxide (NMC).

Each experiment is referred to as (k, x, T_{amb}) , where k refers to the cell sample, x to the applied C-rate, and T_{amb} to the reference temperature. Visualization of voltage, current and Surface Temperature of this dataset is shown in Fig. 2 (b).

3. Dataset 3

For this dataset, a brand new 3Ah LG HG2 battery cell was tested inside an 8 ft³ thermal chamber. The testing was performed using a Digatron Firing Circuits Universal Battery Tester operating at 75 A and 5 V, with a voltage and current accuracy of 0.1% of the full scale [11].

Visualization of voltage, current and Surface Temperature of this dataset is shown in Fig. 2 (c).

B. Data Preprocessing and Evaluation Scenarios

Considering the controllability and stability of the battery charging process, only pure charging data was selected for modeling to enhance the universality and accuracy of the method. Since the measurement data was collected at fixed intervals, the data size varied significantly depending on the battery charging rate and cycle count. Among all data, the maximum number of single-charge time points was approximately 300. To standardize the data, single-charge datasets with fewer than 300 time points were uniformly segmented into 300 points within their respective charging cycles. A linear interpolation algorithm was then applied to accurately estimate the corresponding voltage, current, and charge values, constructing the V , I , and Q vectors for each battery.

This study utilized the NMC dataset to train baseline models, with 25°C, 35°C, and 45°C serving as temperature condition labels to investigate the impact of sample balance on model performance. Different combinations of battery sample quantities under varying temperatures were designed and used to form training sets, with the specific number of sample points for each combination shown in Fig. 3. Specifically, Case1 (sample quantity combination [4637, 4806, 4660]) represented a balanced sample scenario; Case2 ([2432, 4806, 15836]) and Case3 ([5531, 2401, 872]) represented imbalanced sample scenarios.

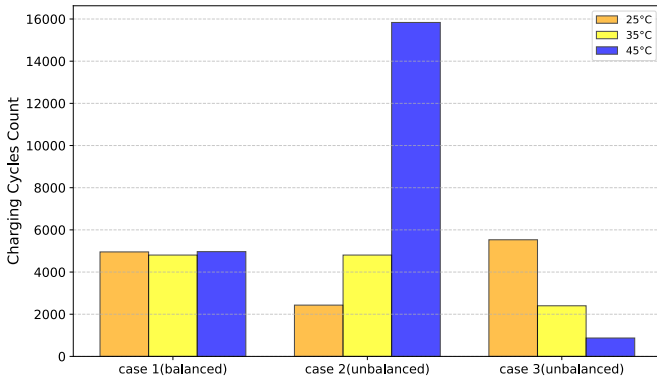


Fig. 3. The combination of battery sample quantities at different temperatures.

The training processes of these datasets under the same baseline model are shown in Fig. 4. We observed that when the sample quantities of temperature labels were imbalanced, the model's training error increased significantly with severe fluctuations. Specifically, an excessively high proportion of samples from one temperature led to the model overfitting to that temperature's characteristics, diminishing generalization capabilities under other temperatures. Conversely, when the sample distribution across temperatures was balanced, the model maintained better equilibrium among different temperature features, resulting in lower training errors and more stable performance.

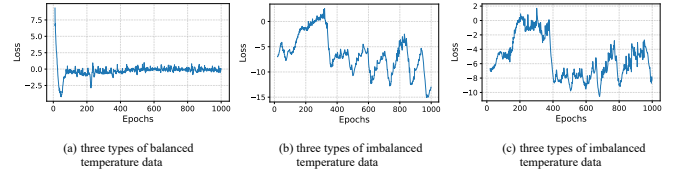


Fig. 4. The training error of the model under different combinations.

C. Experiment Results and Analysis

1) *Assessment of Data Generation Quality*: Take the NMC dataset as an illustration. Fig. 5 (a), (b), and (c) present the comparison between the voltage, current, and capacity curves generated by the model and the real - world data. In these figures, the horizontal axis represents the timestamp of a single charging cycle, while the vertical axis indicates the battery data. Visually, the data generated by this framework is extremely similar to the real data and is almost indistinguishable from it. This serves as a strong testament to the effectiveness of the framework in data generation.

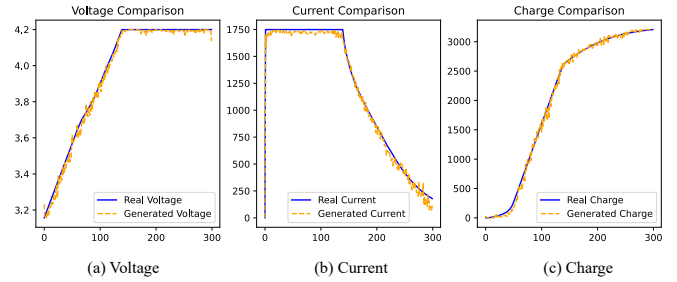


Fig. 5. Comparison between generated data and real data.

The Probability Density Function (PDF) can clearly show how data is distributed across different value ranges. It can be used to determine whether the distribution of the generated data is similar to that of the real data. The experimental results are presented in Fig. 6. Here, Fig. 6 (a), (b), and (c) respectively correspond to the probability distribution curves of the generated data and the real data for Dataset 1, Dataset 2, and Dataset 3. It is evident that the probability distribution curves of the generated data and the real data overlap to a great extent. This fully indicates that there is a significant similarity in the data distribution between the two.

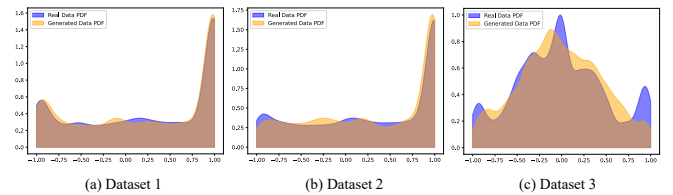


Fig. 6. PDF evaluation of generated data under different datasets.

Principal Component Analysis (PCA) reduces high-dimensional data to lower dimensions, revealing the key characteristics of the data in a low-dimensional space. The distributions of generated and real data in the principal component space are shown in Fig. 7. For Dataset 2, the generated data shows a high degree of consistency with the real data in the two-dimensional principal component space, indicating that the generated data accurately reconstructs the high-dimensional features of the original data. Although the generated data for Datasets 1 and 3 show some sparse points in regions where real data is less dense, their principal component distributions remain highly consistent with the core features of the real data.

Overall, the generated data effectively captures the main distribution patterns of real data in the feature space, verifying the high-fidelity reconstruction capabilities of the generative model for the original data's primary characteristics. The local deviations in Dataset 1 and Dataset 3 may stem from the sparsity of the original data and do not affect the overall validity of the generated data quality.

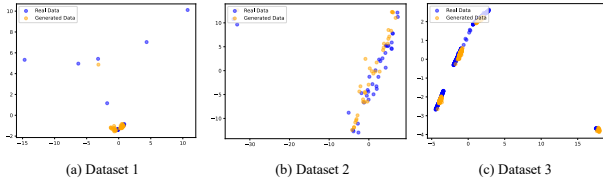


Fig. 7. PCA evaluation of generated data under different datasets.

2) *Battery State Estimation Performance*: To accurately estimate the battery's SOH and SOC, we employed a Bi-GRU architecture for prediction. The Bi-GRU architecture effectively processes time-series data by leveraging bidirectional information flow to capture temporal variations in battery states. Four data utilization schemes were designed:

- Real Data Only: Using solely real-world battery test data.
- Generated Data Only: Relying entirely on synthetic data generated by the proposed algorithm.
- Data Replacement and Repair: Replacing missing or inaccurate real data with generated data to address gaps.
- Data Augmentation: Fusing all real and generated data to enhance data diversity.

Fig. 8 shows the comparison between the SOH calculated from the data generated by the model developed in this study and the SOH calculated from real data. The graph clearly indicates that the data generated during model training has high credibility.

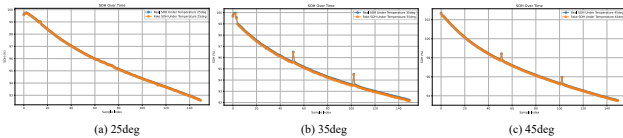


Fig. 8. The variation of SOH for real and generated data at different temperatures.

Taking SOH and SOC prediction as an example, Fig. 9 shows that the data augmentation approach yields predictions closest to real values with the smoothest curves. This improvement stems from the generated data expanding data diversity, compensating for real data limitations. The combination of real and synthetic data provides richer information for the model, enabling better learning of SOH and SOC variation patterns.

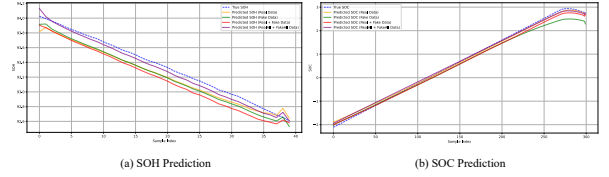


Fig. 9. SOH and SOC prediction during the charging process under different datasets.

Replacing part of the real data with generated data outperforms using only real or generated data, as it preserves key features of the real data while enriching the model's learning material. However, similar to using generated data alone, this method exhibits larger fluctuations in the initial sample points due to the distribution differences between the generated and real battery data, as well as insufficient model adaptation during early training. Furthermore, all three scenarios, except for data augmentation, show significant fluctuations in predictions during the late stages. Using only real data may introduce noise from battery aging and environmental factors, along with cumulative errors in long-term predictions. Relying solely on generated data or replacing real data with it depends on the quality and distribution of the synthetic data, making it hard for the model to capture subtle state changes in complex late-stage battery conditions, leading to significant prediction volatility.

To comprehensively and objectively evaluate model performance, we selected Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Coefficient of Determination (R^2) as evaluation metrics. The results are documented in TABLE I.

Our model, PIT-GAN, excels in data augmentation, achieving the best RMSE (0.0005), MAPE (0.29%), and an almost perfect R^2 (0.9996). This success is due to the hybrid PINN-GRU model, which combines physical laws with dynamic sequences, and the multi-scale convolution, which captures both local and global features.

In comparison, CGAN shows domain - specific adaptability in partial data replacement (RMSE = 0.0042), yet its reliance on clustering labels restricts generalization, and GAN training instability weakens performance in augmentation (R^2 = 0.9890). Transformer-GAN generates good sequences for partial replacement (RMSE = 0.0030), but its pure Transformer-GAN design lacks explicit physical constraint modeling, resulting in a lower R^2 (0.9980) compared to PIT-GAN in augmentation. RNN-GAN struggles with gradient issues in its recursive structure and exhibits a high MAPE (2.54%) in partial replacement, showing poor performance in capturing complex sequences.

TABLE I
EVALUATION RESULTS OF DIFFERENT METHODS IN DIFFERENT DATA SCENARIOS

Evaluation metrics	RMSE			MAPE			R ²		
	Real data	Partial data replacement	Data augmentation	Real data	Partial data replacement	Data augmentation	Real data	Partial data replacement	Data augmentation
PITGAN	0.0021	0.0039	0.0005	1.31%	2.41%	0.29%	0.9916	0.9795	0.9996
CGAN	0.0021	0.0042	0.0009	1.31%	2.65%	0.49%	0.9916	0.9658	0.9890
RNN_GAN	0.0021	0.0042	0.0009	1.31%	2.54%	0.55%	0.9916	0.9755	0.9919
Transformer_GAN	0.0021	0.0030	0.0009	1.31%	2.34%	0.40%	0.9916	0.9774	0.9980

Remarkably, PIT-GAN achieves the lowest RMSE and MAPE across all scenarios, demonstrating the combined power of its physical constraints and dynamic modeling. It effectively addresses the generalization and stability issues found in traditional GANs in the battery field, while ensuring data authenticity.

D. Parameter Analysis

In the experimental data generation framework for lithium batteries based on PIT-GAN, the adjustment of key hyperparameters significantly impacts performance:

1. **Gradient Penalty Coefficient:** It controls the contribution to the loss function, optimizing training stability and preventing gradient anomalies. Excessive values impede discriminator convergence, while too low values weaken the constraint effect.
2. **Learning Rate:** Set separately for the Adam optimizers of the generator and discriminator, it controls parameter update steps, balancing convergence speed and performance. In lithium battery experiments, improper settings can cause convergence issues or prolong training.
3. **Dropout:** By adjusting neuron dropout probability within [0, 1), set at 0.2 in this experiment, it suppresses overfitting and enhances the model's generalization ability for lithium battery data.

We used the grid search method to determine optimal hyperparameters. The relationship between loss, gradient penalty coefficient, and learning rate is shown in Fig. 10. After analysis, we finally adopted a gradient penalty coefficient $\lambda_{gp} = 14$ and learning rate $l_r = 1$, which performed well in experiments and facilitated the effective generation of lithium battery experimental data.

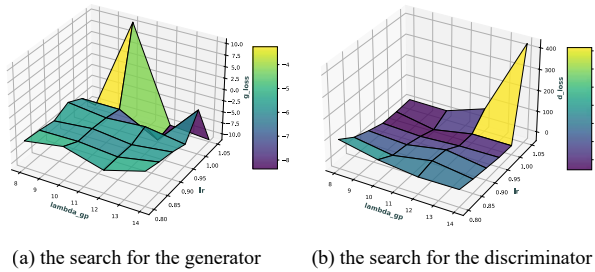


Fig. 10. Grid search results for hyperparameters.

IV. CONCLUSION

This paper proposes the PITGAN model to generate time-series of lithium battery. The model reconstructs voltage, current, and capacity curves. The distribution of the generated data in the PCA-reduced space closely aligns with real data, confirming the high quality of the synthetic time-series data. Additionally, an asymmetric gradient penalty strategy enhances GAN training stability and convergence, achieving optimal RMSE (0.0005) and R² (0.9996) in data augmentation tasks. Furthermore, the model incorporates a weighting coefficient to balance battery samples across different temperature conditions, improving generalization and reducing training errors, particularly in SOC and SOH prediction.

REFERENCES

- [1] A. G. Yeklangi, A. Ghafari, F. A. Sima, S. Akbari, Advancing lithium-ion battery manufacturing: novel technologies and emerging trends, *Journal of Applied Electrochemistry* 54 (12) (2024) 2653–2682.
- [2] M. Sedlářík, P. Vyroubal, D. Capková, E. Omerdic, M. Rae, M. Mačák, M. Šedina, T. Kazda, Advanced machine learning techniques for state-of-health estimation in lithium-ion batteries: A comparative study, *Electrochimica Acta* (2025) 145988.
- [3] S. G. Chae, S. J. Bae, K.-Y. Oh, State-of-health estimation and remaining useful life prediction of lithium-ion batteries using dncnn-cnn, *Journal of Energy Storage* 106 (2025) 114826.
- [4] H. Zhao, C. Zhang, C. Liao, L. Wang, W. Liu, L. Wang, Data-driven strategy: A robust battery anomaly detection method for short circuit fault based on mixed features and autoencoder, *Applied Energy* 382 (2025) 125267.
- [5] S. Liu, D. Xu, Y. Ye, T. Pan, Intelligent fault diagnosis for unbalanced battery data using adversarial domain expansion and enhanced stochastic configuration networks, *Information Sciences* 687 (2025) 121399.
- [6] C. Hu, F. Cheng, Y. Zhao, S. Guo, L. Ma, State of charge estimation for lithium-ion batteries based on data augmentation with generative adversarial network, *Journal of Energy Storage* 80 (2024) 110004.
- [7] B. Iranipour, M. Sadeghian, E. Mohajerani, Artificial data generation: A strategy to improve efficiency predictions in mixed sn-pb perovskite solar cells, *Materials Today Communications* 43 (2025) 111625.
- [8] H. Jin, R. Ru, L. Cai, J. Meng, B. Wang, J. Peng, S. Yang, A synthetic data generation method and evolutionary transformer model for degradation trajectory prediction in lithium-ion batteries, *Applied Energy* 377 (2025) 124629.
- [9] J. Zhu, Y. Wang, Y. Huang, R. B. Gopaluni, Y. Cao, M. Heere, M. J. Mühlbauer, L. Mereacre, H. Dai, X. Liu, A. Senyshyn, X. Wei, M. Knapp, H. Ehrenberg, Data-driven capacity estimation of commercial lithium-ion batteries from voltage relaxation, *Nature Communications* 13 (1) (2022) 2261.
- [10] E. Catenaro, S. Onori, Experimental data of lithium-ion batteries under galvanostatic discharge tests at different rates and temperatures of operation, *Data in Brief* 35 (2021) 106894.
- [11] T. T. D. Nguyen, S. Abada, A. Lecocq, J. Bernard, M. Petit, G. Marlair, S. Grugeon, S. Laruelle, Understanding the thermal runaway of ni-rich lithium-ion batteries, *World Electric Vehicle Journal* 10 (4) (2019).